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Study on Proof-of-Concept for Resource-Optimised AI Benchmarking: Efficiency, Cost, and Sustainability in Small-Medium Enterprises (SMEs)

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written by
Heansuh Lee
Matriculation Number: 944320

First Supervisor: Prof. Dr. Michael Prange
Second Supervisor: Prof. Dr. Stephan Schneider

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Study on Proof-of-Concept for Resource-Optimised AI Benchmarking: Efficiency, Cost, and Sustainability in Small-Medium Enterprises (SMEs)

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Abstract

While Artificial Intelligence (AI)'s capabilities are becoming more powerful than ever, they demand substantial computational resources, raising critical concerns about cost, scalability, and environmental impact, which is already becoming a significant issue, for example with the data centres and Graphics Processing Unit (GPU) clusters. This brings challenges for small and medium-sized enterprises (SMEs), which often lack the infrastructure and capital of their larger counterparts, hindering their ability to adopt AI efficiently. This thesis aims to cover this gap by proposing a Proof-of-Concept (PoC) benchmarking framework designed to assess AI efficiency across four key dimensions: accuracy, runtime, cost, and energy consumption. The framework is developed through a comprehensive literature review of existing benchmarking tools and a critical evaluation of modern AI hardware architectures, including GPUs, specialised accelerators, and emerging data centre infrastructures. As a practical implementation, this research introduces Project HANSAL (Hybrid AI for Next-gen: Sustainable, Affordable, and Lightweight). This concept is applied to the framework to provide SMEs with insights for their domain-specific business cases by measuring their AI systems that strategically balance high performance with resource sustainability. Providing a structured framework and reference guide for resource optimisation in AI, this thesis aims to offer both academic and industrial values by supporting energy-aware, cost-efficient AI adoption. While it focuses on its applicability for SMEs, it touches on the possibility for larger corporations to adopt this concept and framework, based on their business needs and counterparts.

Declaration

I, Heansuh Lee, declare that this thesis is my own work and has not been submitted in any form for another degree or diploma at any university or other institute of tertiary education. Information derived from the published and unpublished work of others has been acknowledged in the text and a list of references is given in the bibliography.

City, Date

Heansuh Lee

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Contents

1 Introduction

Artificial intelligence (AI) systems have advanced rapidly over the past decade, driven in particular by deep learning models with growing parameter counts and training data scales. These advances have led to substantial increases in computational demand, which translate directly into higher electricity consumption and associated carbon emissions from data centres and specialised hardware such as GPU clusters. Recent analyses show that training and operating modern AI models can consume megawatt-hours (MWh) of energy and contribute non-negligibly to the overall climate impact of information and communication technologies.

1.1 Motivation

AI systems, particularly modern deep learning models, require large volumes of computation for both training and inference, which directly translates into substantial electricity consumption and hardware demand. Studies show that training and operating contemporary deep learning models can consume MWh of energy, with runtime energy use scaling strongly with model size, data volume, and chosen hardware. As these workloads migrate from individual servers to large GPU clusters, their aggregate impact becomes significant within the overall information and communication technology (ICT) sector, which already accounts for several percent of global electricity use and around 1–2% of global greenhouse gas emissions.

A growing share of this footprint is concentrated in data centres, whose electricity demand in Europe has been rising steadily and is expected to continue increasing with the expansion of AI and cloud services. Recent scenario analyses project that data-centre energy use and associated emissions could roughly double over the coming decade, driven in part by specialised AI infrastructure and dense GPU farms required for training and serving large models. These trends create tension with climate-policy targets, as uncontrolled growth in AI-related energy demand risks undermining broader decarbonisation efforts in the power sector.

In response, the European Union and its member states have articulated explicit goals to make data centres more energy and resource-efficient, including the objective of climate-neutral data centres by 2030 and planned regulatory packages focused on data-centre energy efficiency. Policy analyses around the European Green Deal and the AI Act highlight that energy and resource efficiency should become a core design objective for AI systems, not an afterthought, if AI is to support rather than hinder the energy transition. Against this backdrop, “green AI” and resource-aware AI operation—where energy use, cost, and emissions are treated as metrics alongside accuracy and latency—are increasingly important, especially for organisations that lack the capacity to absorb uncontrolled increases in computational demand.

1.2 Context

Small and medium-sized enterprises (SMEs) typically operate under tighter financial and organisational constraints than large technology companies, which limits their access to specialised AI infrastructure such as dedicated GPU servers, high-performance storage, and professional MLOps teams. Instead, many SMEs rely on a small number of on-premise machines, shared virtual private servers, or pay-as-you-go cloud instances, making it difficult to experiment broadly with different hardware configurations or large-scale models without incurring prohibitive costs.

At the same time, SMEs increasingly view AI as a key lever for maintaining competitiveness, for example through predictive maintenance, demand forecasting, or customer analytics, and thus face pressure to integrate AI into their products and processes. However, they often lack practical guidance on how different combinations of models, frameworks, hardware, and cloud offerings translate into concrete trade-offs between accuracy, runtime, energy use, and monetary cost for their own workloads. This uncertainty can result in suboptimal choices—either overprovisioning expensive infrastructure or underinvesting in AI capabilities—and creates a barrier to energy-aware, cost-efficient adoption of AI in the SME context.

1.3 Problem Statement

Many established AI benchmarks and leaderboards concentrate primarily on model accuracy, and in some cases latency, while assuming access to substantial computational resources such as multi-GPU servers or large cloud clusters. These benchmarks provide valuable insights for model comparison at scale, but they rarely expose energy consumption, monetary cost, or carbon footprint in a form that smaller organisations with limited hardware and budgets can readily apply to their own environments. As a result, SMEs have little empirical support for deciding whether a given model, framework, or deployment option offers an acceptable balance between performance and resource usage on realistic, resource-constrained setups.

In parallel, several specialised tools have emerged that enable more detailed measurement of AI energy use and emissions, including CodeCarbon for estimating carbon footprint from power and grid-intensity data, as well as Zeus and Perun for fine-grained, hardware-level energy profiling of deep learning and high-performance computing workloads. While powerful, these tools are often presented as independent projects or research prototypes, and there is currently no unified, SME-oriented benchmarking framework that integrates them into a coherent workflow which jointly reports accuracy, runtime, cost, and energy consumption for typical AI workloads. This gap makes it difficult for SMEs to systematically compare alternative AI configurations and to incorporate energy- and cost-awareness into everyday model and infrastructure decisions.

1.4 Research Objectives

This thesis develops and evaluates a Proof-of-Concept (PoC) benchmarking framework, Project HANSAL, that enables systematic assessment of AI workloads along four key dimensions: accuracy, runtime, cost, and energy consumption on realisti-

cally constrained hardware. The goal is to provide SMEs with a practical, reproducible basis for making informed decisions about AI models, infrastructure, and deployment options under resource and budget limitations.

The research is guided with the following questions:

1. How can SMEs systematically compare AI workloads across accuracy, runtime, cost, and energy consumption using a unified benchmarking framework, and what are the key design considerations?

2. To what extent do existing measurement tools support reliable and interpretable comparisons of AI workloads on modest, SME-typical hardware and infrastructure? What is the current state-of-the-art, and where are the gaps?

3. In terms of granularity and metrics, what levels of detail are expected and useful for SMEs when evaluating AI workloads for energy and cost efficiency?

4. What trade-offs between model performance and resource usage emerge in representative workloads when evaluated with this framework, and how can these trade-offs inform energy-aware, cost-efficient AI adoption in SMEs?

2 Background and Literature Review

This chapter provides foundational context for energy-aware AI benchmarking in containerised environments, surveying key concepts, metrics, and tools while identifying gaps that this work addresses. The subsequent sections synthesize fundamentals of AI benchmarking, energy/CO_2 metrics, containerization challenges, and tools such as CodeCarbon, Perun, and Kepler, organized thematically to highlight methodological trends, accuracy debates, and unresolved issues in multi-tenant settings. Each section critiques pivotal works and links directly to the thesis research questions on scalable, container-level energy accounting.

2.1 Energy Consumption

Global energy consumption patterns exhibit steady growth, driven by population expansion, industrialisation, and digitalisation, with electricity demand rising at ∼2–3% annually to reach approximately 28 000 TWh in 2024. Renewables now supply over 30% of this demand, yet fossil fuels still dominate at ∼60% [1]. Data centres exemplify this trend, consuming 1–2% of global electricity (415 TWh in 2024) and projected to double to 900–1 000 TWh by 2030, propelled by AI's exponential compute requirements [2][3].

2.1.1 Data Centre Energy Demand Scale

Global data centre electricity use reached around 415 TWh in 2024, equivalent to the annual consumption of countries like Sweden or Australia, with AI training and inference expected to comprise 35–50% of this by 2030. Projections vary due to model assumptions, but the International Energy Agency forecasts more than doubling to 945 TWh by 2030, underscoring the need for granular measurement to inform grid planning. Per-facility demand can exceed 100 MW for hyperscale sites, rivaling small cities and influencing regional carbon intensity based on grid mix.

Electricity demand follows distinct seasonal and diurnal cycles, peaking in summer/winter due to cooling/heating loads, while data centres contribute baseload patterns that flatten daily profiles but introduce surges during AI training. Post-2020 trends show renewables growing at 10%+ annually, yet AI/data centre expansion could offset 20–50% of efficiency gains without targeted interventions. In Europe and the US, data centre demand now rivals aviation emissions (∼2% of global CO_2).

2.1.2 Data Centre Energy and Water Demand Patterns

Global electricity demand reached approximately 28 000 TWh in 2024, growing at 2–3% annually due to industrialisation and digitalisation, with distinct diur-

nal/seasonal patterns: baseload from data centres flattens daily profiles but introduces AI-driven surges during training/inference peaks, exacerbated by summer cooling loads in warm climates. Renewables supplied over 30% but fossil fuels dominated at ∼60%, with data centres consuming 415 TWh (1–2% of global total) and projected to double to 900–1 000 TWh by 2030.

Hyperscalers (Microsoft, AWS, Google, Meta) drive this via massive CapEx: $290B total in 2024 (40%+ growth expected 2025), building GW-scale campuses with 220 GW targeted capacity by 2030; e.g., Microsoft/Google each announced 100+ new facilities, adding 10s of GW via liquid-cooled AI racks. Water usage parallels energy, with data centres withdrawing 1–2 billion m^3/year globally (equivalent to 500,000 Olympic pools), primarily for evaporative cooling; WUE averages 1.8 L/kWh but hyperscalers target <1.0 via air/closed-loop systems, though AI heat densities strain this (e.g., 100 MW sites need 200 ML/day).

ESG frameworks (GRI 302/305, SASB TC-SI-130a) mandate reporting these via PUE/WUE/carbon intensity for investor transparency, linking facility demand to tenant AI workloads. Key environmental indicators for ESG's 'E' (Environmental) pillar include energy consumption (Scope 1/2 emissions), greenhouse gas emissions (CO2e), water usage (WUE), waste generation, and biodiversity impact, often reported via frameworks like GRI (Global Reporting Initiative), SASB (Sustainability Accounting Standards Board), and TCFD (Task Force on Climate-related Financial Disclosures). These link directly to data centres/AI through metrics like PUE (energy efficiency), WUE (liters/kWh), and carbon intensity (gCO2/kWh), with standards such as ISO 14064 (GHG accounting) and EU CSRD mandating disclosures for high-impact sectors including hyperscalers.

2.1.3 Measurement Methods in Data Centres

Energy consumption in data centres follows a hierarchical measurement approach, starting from high-level facility totals down to granular IT equipment and tenant-specific allocations. The primary metric, Power Usage Effectiveness (PUE), quantifies overall efficiency by dividing total facility energy (IT equipment + cooling + power delivery + lighting) by IT equipment energy alone, with industry averages of 1.5–1.7 and leading hyperscalers targeting below 1.1 through liquid cooling and free-air systems.

Rack-level metering provides the foundational data, deploying redundant power distribution units (PDUs) with kW sensors per outlet or circuit, capturing real-time draw at 1–15 minute granularity for billing and capacity planning. These measurements aggregate upward: individual server power comes from baseboard management controllers (IPMI for legacy, Redfish for modern) querying CPU/GPU/DRAM via hardware counters like Intel RAPL or NVIDIA NVML, while building management systems (BMS) track HVAC, UPS efficiency, and lighting via thousands of sensors fused into facility PUE.

Per-tenant allocation occurs in colocation environments through dedicated PDUs per customer rack or cage, enabling billed kW commitments (e.g., 5–50 kW/rack) adjusted by measured overages. Multi-tenant clouds further disaggregate via virtual meters tied to VM/container usage, combining hypervisor counters with workload logs for approximate per-customer kWh. Cooling overhead, comprising 30–50% of total energy, gets estimated via psychrometric models or direct flow/temperature

sensors, though dynamic AI workloads challenge static PUE due to varying heat densities.

This layered approach supports ESG reporting but reveals gaps at container/AI workload granularity, motivating tools like Kepler for Kubernetes-native accounting that bridges facility meters with per-pod energy via eBPF and cGroup metrics.

2.1.4 Importance of Per-Company and Per-User Allocation

Operators allocate demand to companies (tenants) for billing, sustainability reporting, and efficiency incentives; for instance, colocation providers charge based on metered rack kW, adjusted by PUE to include overhead. End-user allocation (e.g., per AI query) supports product-level carbon footprints, enabling users to optimize prompts or models for lower emissions, as seen in emerging cloud APIs reporting kWh per inference. Without such breakdowns, externalities like unpriced emissions persist, delaying net-zero transitions.

2.2 AI Benchmarking

AI benchmarking evaluates model performance across metrics like accuracy, latency, throughput, and resource efficiency. Standardized suites (e.g., MLPerf, GLUE, ImageNet) provide consistent comparisons, while real-world workloads assess practical applicability. Emerging benchmarks focus on energy efficiency and carbon footprint, reflecting growing environmental concerns. Key challenges include ensuring reproducibility, accounting for hardware variability, and balancing performance with sustainability.

2.2.1 Existing AI Benchmarks

Numerous AI benchmarks exist to evaluate model performance across various tasks and metrics. Some of the most prominent benchmarks include:

- **MLPerf**: Industry-standard suite from MLCommons covering training, inference, and storage across vision, NLP, and recommendation tasks on diverse hardware (CPU/GPU/TPU). Includes power walls measuring perf/watt and total energy alongside traditional latency/throughput metrics.
- **ImageNet**: Foundational computer vision benchmark with 1.2M labeled images across 1000 classes, evaluating classification accuracy and transfer learning baselines. ILSVRC challenges established CNN architectures like ResNet and EfficientNet.
- **GLUE/SuperGLUE**: NLP leaderboards testing general language understanding via tasks like sentiment analysis, textual entailment, and question answering. Measures average score across 9–28 datasets, driving BERT/RoBERTa/T5 development.
- **SQuAD**: Reading comprehension benchmark requiring extractive answers from Wikipedia passages, emphasizing information retrieval and contextual understanding in transformer models.

- **MLCommons Tiny**: Embedded/ML edge benchmark for keyword spotting, anomaly detection, and image classification on microcontrollers, balancing accuracy against memory footprint and inference latency.
- **Big-Bench/HELM**: Massive canonical/multi-task collections probing reasoning, ethics, and robustness across 200+ tasks, resistant to overfitting via held-out evaluations.
- **DAWNBench/Time-to-Train**: End-to-end training time/cost minimization challenges, measuring wall-clock time to target accuracy on CIFAR/ImageNet rather than FLOPs alone.

These benchmarks reveal trade-offs: MLPerf excels in industrial reproducibility but hardware-specific submissions limit cross-platform energy comparisons, while academic suites like GLUE prioritize task diversity over power measurement.

2.2.2 Algorithm Measurement Technologies

State-of-the-art energy measurement for AI algorithms combines hardware counters, software profilers, and ML-specific frameworks to capture power draw from CPU, GPU, and system-level components during training and inference.

Hardware counters provide the most accurate direct measurements:

- **RAPL** (Intel/AMD): Package-level energy for CPU, DRAM, and platform (uncore), accurate to 1–5% at sub-second granularity via model-specific registers (MSRs).
- **NVML/DCGM** (NVIDIA): Real-time GPU power, memory bandwidth, and tensor core utilization across A100/H100 generations.
- **ROCm SMI** (AMD): MI200/MI300X GPU power and fabric metrics for open-source alternatives.

Software profilers enable detailed analysis:

- **Intel VTune**: CPU hotspot profiling with power attribution to functions/threads via PMU events.
- **NVIDIA Nsight Systems/Compute**: Kernel-level GPU timeline with energy-per-operation breakdown.
- **perf**: Linux tool exposing PMU counters for custom energy models across x86/ARM.

ML-specific frameworks bridge FLOPs to energy:

- **MLPerf Power Walls**: Measures perf/watt and total kWh alongside latency/throughput in standardized training/inference submissions.
- **TensorFlow/PyTorch Profilers**: Hardware FLOPs counting converted to energy via pre-measured power curves (10–20% estimation error).

These technologies enable per-epoch power logging and hardware-agnostic comparisons, though integration challenges persist in containerized/multi-tenant environments addressed by subsequent sections.

2.3 Code-Level Energy Measurement Tools

Code-level tools enable fine-grained energy tracking directly within Python/ML workflows, wrapping training loops and inference functions to log power draw, carbon estimates, and efficiency metrics without requiring infrastructure changes.

2.3.1 CodeCarbon

CodeCarbon provides a lightweight Python decorator for automatic energy and carbon tracking across local and cloud environments. Installation occurs via `pip install codecarbon`, followed by simple function wrapping:

```
from codecarbon import OfflineEmissionsTracker
tracker = OfflineEmissionsTracker(country="DE", region="Nord")
tracker.start()
# Training code here
tracker.stop()
```

The tool queries RAPL/NVML for hardware power, multiplies by CPU time and regional grid carbon intensity (gCO_2/kWh), and outputs CSV logs with kWh, CO_2e, and monetary estimates. Cloud mode estimates via provider PUE/carbon factors when hardware counters are unavailable. Limitations include single-process focus and CPU-heavy estimation bias on GPU workloads.

2.3.2 Zeus

Zeus extends MLPerf benchmarking with comprehensive power analysis, capturing GPU/CPU energy across full training runs with statistical aggregation. Launched via CLI on standard MLPerf benchmarks:

```
zeus benchmark mlperf/training/dlr/resnet50 --duration 3600 --power
```

It logs per-epoch power, temperature, and throughput with error bars from multiple runs, enabling perf/watt comparisons across hardware generations. Integration with MLPerf submission pipelines supports industrial validation, though setup requires benchmark-specific containers and NVML access.

2.3.3 Perun

Perun, developed by Polish researchers, offers fine-grained CPU/GPU monitoring via Linux perf events and ROCm integration, designed for research pipelines. Configuration targets specific workloads:

```
perun monitor --gpu --output results.json python train.py
```

Outputs include JSON/CSV with power traces, FLOPs attribution, and custom metrics for data engineering workflows. Multi-GPU support and container compatibility via Docker exec make it suitable for Kubernetes experimentation, though documentation remains primarily Polish/English academic papers.

These tools bridge application code with hardware telemetry but face challenges in multi-tenant attribution and real-time grid intensity updates addressed by container orchestrators in subsequent sections.

2.4 Container-Level Energy Measurement

Container-level energy measurement addresses the gap between code-level profilers and facility-wide metering by attributing power draw to individual pods and containers running in Kubernetes or Docker environments. Tools like Kepler bridge hardware telemetry with orchestrator metrics, enabling per-workload accounting in multi-tenant deployments.

2.4.1 Kepler

Kepler (Kubernetes-based Efficient Power Level Exporter) provides container/pod-level energy attribution through eBPF probes and cAdvisor integration, exporting comprehensive metrics to Prometheus for real-time visualization. Deployment follows standard Kubernetes patterns:

```
kubectl apply -f https://github.com/sustainable-computing-io/\\
kepler/releases/download/v1.0.0/kepler.yaml
```

Core measurement combines:

- eBPF: Kernel-level CPU/memory usage from cGroups v2.
- cAdvisor: Container runtime metrics (Docker/CRI-O).
- Hardware Exporters: RAPL/NVML/AMD SMI.
- Energy Models: Pod fractions mapped to socket power.

PromQL query examples:

```
sum(rate(container_energy_joules_total
{namespace="ai-bench"}[5m])) by (pod)
```

Key advantages include multi-tenant isolation, GPU support (NVIDIA/AMD), and 5–10% accuracy vs rack meters.

2.4.2 Complementary Approaches

Additional tools:

- Kubelet Metrics: Native cGroup stats → RAPL ratios.
- NVIDIA DCGM-Exporter: GPU-only power per container.
- Custom Sidecars: CodeCarbon/Perun in init containers.

These enable experiments comparing container vs code-level measurements across Docker/Kubernetes, validating attribution accuracy for green AI benchmarking.

2.5 Standards and Reporting Frameworks

Standardization efforts for AI energy measurement and sustainability lag behind traditional IT infrastructure, with fragmented initiatives targeting transparency, comparability, and regulatory compliance across algorithm, container, and system levels.

2.5.1 ISO/IEC Standards

The ISO/IEC 30148 series (AI sustainability and environmental impact) provides the most comprehensive framework, currently at draft stage (TR 30148–1:2022). Key components include:

- **30148–2**: AI system lifecycle carbon footprint, specifying energy accounting boundaries (training, inference, retraining).
- **30148–3**: Metrics for transparency (FLOPs, kWh, CO_2e per task) and efficiency benchmarking.
- **30148–5**: Containerized deployment guidance, addressing orchestration overhead attribution.

These establish terminology (direct/indirect emissions, static/dynamic measurement) but remain voluntary, with full publication expected 2026+.

2.5.2 IEEE and Industry Standards

IEEE P3109 (AI Carbon Footprint) proposes model cards including measured/estimated energy across hardware configurations, complementing MLPerf power walls with reproducibility requirements. The Green Software Foundation's Software Carbon Intensity (SCI) metric quantifies software impact via:

$$SCI = E \times I \times P$$

where E = energy consumed, I = carbon intensity (gCO_2/kWh), P = performance penalty factor. SCI enables container-level comparisons across clouds.

2.5.3 Regulatory Frameworks

The EU AI Act (effective 2026) mandates high-risk AI systems (>10M parameters or critical applications) report training energy and carbon, verified by third-party auditors. California's SB-942 requires data centre operators disclose PUE/WUE annually, while voluntary codes (EU Code of Conduct for Data Centres) target PUE<1.3 by 2027.

2.5.4 Gaps and Thesis Relevance

No binding algorithm/container standard exists; current frameworks emphasize facility-level reporting over workload granularity. ISO/IEC 30148 lacks container-specific guidance, IEEE P3109 omits multi-tenant attribution, and SCI assumes static hardware power profiles inadequate for dynamic AI workloads. This thesis addresses these gaps through scalable container-level benchmarking aligned with emerging standards.

2.6 AI Benchmarking Systems

AI benchmarking systems standardize evaluation across hardware ecosystems, evolving from compute-centric metrics toward comprehensive sustainability assessment including energy efficiency, power walls, and carbon-aware performance.

2.6.1 MLPerf (MLCommons)

MLPerf, developed by MLCommons consortium (Google, NVIDIA, Intel, AMD, hyperscalers), represents the industry gold standard for reproducible ML benchmarking. Core branches include:

- **MLPerf Training**: End-to-end time-to-train for ResNet-50, BERT, GPT-3 on massive datasets, submitted across 100+ systems yearly.
- **MLPerf Inference**: Closed/open divisions measuring throughput/latency at fixed accuracy (e.g., 99% ImageNet).
- **MLPerf Storage/Inference Edge/Tiny**: I/O bottlenecks, mobile/embedded constraints.

Energy integration via "Power Results" logs total kWh, peak power, and perf/watt using RAPL/NVML during submissions. 2025 v5.0 mandates power walls (max power sustained during convergence), enabling hardware efficiency rankings.

2.6.2 MLPerf Inference Efficiency

The Inference v4.0+ Efficiency Track awards systems maximizing samples/second per watt, directly addressing datacentre economics. Offline/server scenarios test datacentre GPUs, while edge emphasizes battery life. Results reveal 10–100x efficiency gaps across vendors, driving liquid cooling and sparsity optimizations.

2.6.3 Complementary Systems

- **DAWNBench**: Time/cost-to-target accuracy, emphasizing total dollars (energy proxy).
- **FLOPs Benchmarks**: Theoretical compute (HuggingFace Open LLM Leaderboard) converted to energy via hardware models.
- **CMLPerf (Cloud MLPerf)**: Container-orchestrated extensions for Kubernetes, measuring orchestration overhead.

2.6.4 Limitations and Extensions

MLPerf excels in reproducibility but focuses closed submissions on flagship hardware, limiting open-source applicability. Energy metrics assume stable power profiles inadequate for bursty inference; no multi-tenant or carbon intensity factors. This thesis extends MLPerf-style protocols to containerized green AI benchmarking, incorporating grid-aware scheduling and per-tenant attribution missing from current systems.

2.7 Efforts to Measure AI Energy Consumption

Recent initiatives systematically measure AI energy across research, industry, and regulatory domains, enabling model optimization, hardware selection, and emissions reporting amid growing datacentre demands.

2.7.1 Research and Benchmarking Efforts

MLPerf's "Power Results" extension captures energy walls (peak/average power during convergence), perf/watt ratios, and total kWh across 50+ submissions annually, revealing efficiency leaders (H100 > A100 by 3x). PapersWithCode integrates CodeCarbon/Experiment Impact Tracker, requiring energy logs for leaderboard entries—BERT fine-tuning reports span 0.1–10 kWh depending on quantization and batch size.

2.7.2 Cloud Provider APIs

Hyperscalers expose per-inference/train-job metrics:

- **Azure ML**: Carbon tracker API returns kWh/CO_2e per endpoint call, factoring regional PUE/grid mix.
- **GCP Vertex AI**: Emissions dashboard logs GPU-seconds → energy with 24/7 carbon-free percentage.
- **AWS SageMaker**: Processor metrics + estimator yielding gCO_2/1000 inferences.

Accuracy relies on internal telemetry (95%+ claimed), though boundary definitions vary (IT-only vs full PUE).

2.7.3 Academic and Open-Source Initiatives

- **Green Algorithms**: Curates 100+ model energy comparisons, finding Llama-7B (15 kWh training) < GPT-3 (1,287 MWh).
- **Carbontracker/experiment-impact-tracker**: GitHub repos with 10k+ stars, automating MLflow integration for reproducible footprints.
- **ML Energy Leaderboard**: Stanford-curated database benchmarking 200+ models across hardware.

2.7.4 Applications and Impacts

- **Model Selection**: DistilBERT saves 40% energy vs BERT at 3% accuracy drop, adopted by HuggingFace defaults.
- **Green AI Research**: Guides quantization/pruning—QLoRA reduces fine-tuning by 90% energy.
- **Regulatory Compliance**: EU AI Act documentation, SEC climate disclosures for AI-heavy firms.
- **Carbon-Aware Scheduling**: Shift inference to renewables (Google DeepMind reduced emissions 30%).

These efforts establish baselines but lack container-orchestration integration and multi-tenant granularity central to this thesis's contributions.

2.8 AI Chip Development and Datacenter Evolution

AI accelerators dominate datacentre compute, powering 90%+ of training/inference workloads with rapid generational improvements doubling perf/watt biennially. Edge/on-device chips increasingly handle inference, threatening centralized datacentre economics while Korean/global startups challenge NVIDIA dominance.

2.8.1 Datacenter AI Chips (NVIDIA Dominance)

NVIDIA commands 80–90% market share through Hopper-to-Blackwell progression, optimizing massive clusters via HBM memory and NVLink interconnects.

Table 2.1: NVIDIA Datacenter GPU Generations (H100 → Blackwell)

Chip	Memory	BW	TDP	Status
H100	80GB HBM3	3.4 TB/s	700W	Mature
H200	141GB HBM3e	4.8 TB/s	700W	Deployed
B200	192GB HBM3e	8 TB/s	1kW	2025

Progress stems from TSMC process shrinks (4NP), FP4/FP6 precision, and sparsity acceleration.

2.8.2 Global Challengers and Korean Innovation

Korea invests heavily in inference chips via Samsung/Rebellions, while US startups target training alternatives:

Table 2.2: AI Chip Challengers

Company	Chip	Status
Rebellions (KR)	Rebel NPU	Prod. 2025
Groq	LPU	$2.8B clusters
Cerebras	WSE-3	Shipping
Tenstorrent	n300	2025
SambaNova	SN40L	Enterprise

Rebellions, Samsung-backed, positions as "Korean NVIDIA" for datacentre/edge inference.

2.8.3 On-Device/Edge Chips: Datacenter Threat

Edge AI shifts 50–70% inference to devices (latency, privacy, cost), growing to $100B by 2030 while training remains centralized.

Edge reduces datacentre traffic 80% for inference (80% AI revenue), commoditizing cloud via Groq/Rebellions racks. Training (trillion parameters) stays exaFLOPS-scale.

Table 2.3: On-Device vs Datacenter AI Chips

Chip	TOPS	Power
Snapdragon X	45	<10W
Apple M4	38	<5W
SiMa MLSoC	50+	<1W
Tesla FSD	100+	Vehicle

2.8.4 Thesis Implications

Benchmarking must evolve with H100→B200 migration, edge inference shift, and multi-vendor chips. Container tools (Kepler) measure hybrid deployments, capturing orchestration overhead absent in bare-metal MLPerf while enabling carbon-aware scheduling across chip generations.

2.9 Summary

This chapter surveyed the energy landscape of AI from global electricity patterns (28 000 TWh, 2–3% annual growth) through data centre scale (415 TWh 2024 → 900–1 000 TWh 2030) to granular container-level measurement, revealing critical gaps in multi-tenant attribution that this thesis addresses. Hierarchical facility metering (PDU → IPMI/Redfish → RAPL/NVML) supports ESG reporting via PUE/WUE but lacks workload granularity, while code-level tools (CodeCarbon, Zeus, Perun) excel in single-process tracking yet fail in orchestrated environments.

AI benchmarking has matured from compute-centric (MLPerf Training/Inference, ImageNet, GLUE) to energy-aware (perf/watt walls, MLPerf Efficiency Track), complemented by cloud APIs (Azure/GCP carbon trackers) and research initiatives (Green Algorithms, ML Energy Leaderboard). Standards progress (ISO/IEC 30148 drafts, IEEE P3109, EU AI Act) establishes terminology but remains voluntary and facility-focused, omitting container orchestration overhead and dynamic carbon intensity.

Key research gaps motivating this work include:

1. **Multi-tenant Attribution**: No standard method exists to allocate shared rack power to individual containers/pods in Kubernetes, essential for cloud billing and tenant sustainability reporting.

2. **Container Orchestration Overhead**: Current tools measure IT energy but exclude scheduler, networking, and storage overheads comprising 10–30% of total draw.

3. **Real-time Carbon Awareness**: Static grid intensity assumptions ignore diurnal/seasonal variations critical for scheduling inference to renewable peaks.

4. **Benchmarking Standardization**: MLPerf excels in bare-metal reproducibility but lacks container-native protocols for cloud-native AI deployments.

This thesis bridges these gaps through scalable container-level energy accounting (Kepler and custom orchestration metrics), validating against facility meters while

extending MLPerf methodologies to Kubernetes. The approach enables per-tenant carbon footprints, carbon-aware scheduling, and compliance with emerging ISO/IEC 30148 requirements, positioning containerized green AI benchmarking as the next evolution in sustainable ML systems engineering.

3 Proof-of-Concept Benchmarking Framework

3.1 Introduction to MLOps and Modern Practices

Machine Learning Operations (MLOps) extends DevOps to ML workflows, automating model training, validation, deployment, monitoring, and retraining in production while ensuring reproducibility, compliance, and cost control. Unlike traditional software, ML introduces data drift, model staleness, and resource variability, necessitating specialized tooling for end-to-end lifecycle management. Modern MLOps in SMEs emphasizes cloud-agnostic, open-source stacks leveraging GitHub Actions for CI/CD, containerization for portability, and observability for sustainability—mirroring Fortune 500 practices at minimal cost.

3.1.1 Current MLOps Landscape

Leading tools include:

- **CI/CD**: GitHub Actions (free tier), GitLab CI, Jenkins—trigger builds/tests on PRs.
- **Orchestration**: Kubeflow, MLflow, ZenML—pipeline DAGs for experiment tracking.
- **Deployment**: Kubernetes (EKS/GKE/AKS), SeldonCore, KServe—inference serving.
- **Monitoring**: Prometheus/Grafana, Weights & Biases—drift detection, metrics.

3.1.2 Cloud Cost Management Tools

MLOps cost overruns (GPUs $10+/hr) demand specialized governance:

- **Kubecost**: Kubernetes cost allocation per namespace/pod, forecasts $/month [Kubernetes native].
- **Cast.ai**: Auto-scales/rightsizes clusters, claims 50–90% savings via spot instances.
- **CloudHealth/Spot.io**: Multi-cloud GPU optimization, interruptible workloads.
- **Kepler + PromQL**: Custom energy-to-cost via $/kWh rates, aligns with thesis focus.

These prevent 30–50% waste from idle GPUs/overprovisioning, essential for SME viability.

3.1.3 Structure Overview

3.2 Monorepo Architecture and Repository Structure

Monorepos consolidate codebases under single Git repositories, enabling atomic changes, shared tooling, and consistent builds across teams—a pattern adopted by Google (entire company), Meta (1.5M files), Microsoft (Windows+Office), Uber (200+ services), and Twitter. SMEs adapt via GitHub for 10–50 use-cases without enterprise overhead.

3.2.1 Monorepo Benefits (Industry Examples)

- **Google**: Single repo for 2B lines, Bazel builds Android/Chrome/Search atomically.
- **Meta**: Shared OSS tools across React/Instagram/PyTorch.
- **Uber**: 1000+ services, Nx for ML/infra consistency.
- **SME Fit**: GitHub Actions matrices parallelize butterfly-vision + future pipelines.

Refactors (PyTorch upgrade) touch one PR vs 10 repos; energy tooling shared across use-cases.

3.2.2 PoC Structure

Root repository: mlops-green-ai-poc

README.md Pipeline setup, quickstart

.github/workflows/ CI/CD YAMLs (build/test/deploy)

docker/ Base Python/ML images

energy-tools/ CodeCarbon/Zeus/Perun wrappers

use-cases/butterfly-vision/ Image classification

monitoring/ Kepler/Grafana dashboards

infra/ SSH cluster scripts

3.2.3 Per-Use-Case YAML Configuration

Each use-case contains a `config.yaml` driving reproducibility and energy compliance:

```
model: resnet50
batch-size: 32
epochs: 50
energy-budget-kwh: 0.5
hardware: local-loq
grid-carbon-gco2-kwh: 250
```

Key fields:

- `model`: Architecture (resnet50, efficientnet)
- `energy-budget-kwh`: CI/CD fail threshold
- `hardware`: local-loq or ssh-cluster
- `grid-carbon-gco2-kwh`: Regional factor (DE: 250g)

3.2.4 Git Workflow and Agile Practices

Branching strategy follows Git Flow for agile MLOps:

- `main`: Production deployments
- `dev`: Staging/integration
- `feature/butterfly-v2`: Parallel development

PRs trigger full validation:

1. Lint (black/mypy) → pass
2. Unit tests (pytest) → 90% coverage
3. Energy benchmark → < config budget
4. Container build → GHCR push
5. Deploy to dev cluster

3.2.5 Agile Scaling Benefits

Monorepo scales effortlessly:

- Add `use-cases/medical-imaging/` → auto-discovered by CI matrix
- Shared `docker/base` → consistent PyTorch/CUDA
- Energy tooling uniform across 10+ pipelines
- GitHub Actions parallelizes → 5min/use-case

Matches FAANG velocity (Google monorepo → 15k commits/day) at SME cost (free tier). Energy gates ensure green compliance from sprint 1.

3.3 CI/CD Pipeline with GitHub Actions

GitHub Actions delivers free CI/CD (2000 min/month) integrated with monorepo, automating lint/test/build/deploy across use-cases via matrix strategies—SME equivalent to GitLab Premium.

3.3.1 Pipeline Stages

Single workflow (`.github/workflows/mlops-pipeline.yml`) triggers on push/PR:

1. **Lint/Test**: black/mypy/pytest (90% coverage)
2. **Energy Benchmark**: CodeCarbon-wrapped train.py vs budget
3. **Container Build**: Docker per use-case → GHCR
4. **Deploy**: kubectl/SSH to dev/prod clusters

3.3.2 Workflow YAML (Key Excerpts)

```
name: MLOps Pipeline
on: [push, pull_request]
jobs:
  matrix-test:
    runs-on: ubuntu-latest
    strategy:
      matrix:
        use-case: [butterfly-vision, medical-imaging]
    steps:
    - uses: actions/checkout@v4
    - run: |
        cd use-cases/${{ matrix.use-case }}
        pip install -r requirements.txt
        black --check .
        pytest --cov=80 tests/
  energy-bench:
    needs: matrix-test
    runs-on: ubuntu-latest
    steps:
    - run: |
        cd use-cases/butterfly-vision
        python src/train.py  # Auto-logs CodeCarbon
    - uses: actions/upload-artifact@v4
      with: {name: energy-report, path: codecarbon.csv}
  deploy:
    needs: energy-bench
    if: github.ref == 'refs/heads/dev'
    runs-on: ubuntu-latest
    steps:
    - name: Deploy to Cluster
```

```
uses: appleboy/ssh-action@v1.0.0
with:
  host: ${{ secrets.CLUSTER_HOST }}
  key: ${{ secrets.SSH_KEY }}
  script: |
    kubectl apply -f use-cases/butterfly-vision/k8s/
```

3.3.3 Secrets and Triggers

GitHub Secrets: `CLUSTER_HOST`, `SSH_KEY`, `GHCR_TOKEN`.
Triggers:

- PR to dev → full pipeline + staging deploy
- Push to main → prod deploy (manual approval)
- Feature branches → test/energy only

3.3.4 SME Advantages

- Parallel matrix → 10 use-cases in 8min
- Artifacts persist energy CSVs for Grafana
- Free caching → 50% faster rebuilds
- SSH-action enables hybrid local/cluster

Energy fail-fast prevents wasteful deploys; badges show pipeline status on README.

3.4 Energy Measurement Tools Integration

Three complementary tools provide hierarchical energy observability: code-level decorators (CodeCarbon/Zeus/Perun) for per-experiment tracking, aggregated by Kepler at container/pod level for MLOps dashboards.

3.4.1 Code-Level Tools Comparison

Table 3.1: Code-Level Energy Tools

Tool	Strengths	Limitations
CodeCarbon	Easy RAPL/carbon	CPU bias
Zeus	Epoch stats	Benchmark-only
Perun	Fine traces	Complex setup

Integration Strategy:

- **CodeCarbon**: Default decorator on `train.py` (CSV to MLflow)
- **Zeus**: MLPerf validation (resnet50 benchmark)

- **Perun**: Custom perf traces

Usage example (`use-cases/butterfly-vision/src/train.py`):

```
from codecarbon import EmissionsTracker
tracker = EmissionsTracker()
tracker.start()
# PyTorch training loop
tracker.stop()
# Fails CI if > config.yaml budget
```

3.4.2 Container-Level: Kepler

Kepler DaemonSet exports pod energy to Prometheus:

```
kubectl apply -f monitoring/kepler.yaml
```

Dashboard query: `sum (container_energy{namespace='mlops'}) by (pod)` visualizes butterfly-vision vs future pipelines.

3.4.3 CI/CD Energy Gating

GitHub Actions sequence:

1. CodeCarbon: Run training, fail if > 0.5 kWh (config.yaml)
2. Artifact: Upload CSV to GitHub for Grafana
3. Kepler: Deploy container, validate overhead vs baseline

Multi-tool approach captures code efficiency and orchestration cost, enabling perf/kWh optimization across use-cases.

3.5 Butterfly Dataset and ML Pipeline

The Butterfly image classification pipeline serves as representative CV workload, enabling end-to-end MLOps validation from data ingestion through energy-monitored serving.

3.5.1 Dataset Description

Butterfly Dataset (Kaggle: 50k images, 5 classes: Monarch, Swallowtail, etc.) mimics production defect detection:

- Train: 40k images (224×224 JPEG)
- Val/Test: 5k each
- Augmentation: Rotation/flip for robustness
- Size: 4GB → fits local LoQ, scales to cluster

Pipeline: preprocess → train ResNet50 → export ONNX → FastAPI serving.

3.5.2 ML Pipeline (`use-cases/butterfly-vision/src/train.py`)

1. **Data**: PyTorch DataLoader with GPU prefetch, augmentation
2. **Train**: CodeCarbon-wrapped ResNet50 (50 epochs, AdamW)
3. **Evaluate**: Val accuracy > 85%, perf/kWh logged
4. **Export**: TorchScript/ONNX artifact
5. **Serve**: FastAPI /predict (batch 1/32)

Hyperparameters from config.yaml; early-stop if energy budget exceeded.

3.5.3 Hardware Setup

Dual-environment deployment validates local-to-cluster portability:

- **Local**: Lenovo LOQ 15IAX9 (i5–12450HX, 16GB DDR5, 1TB NVMe, integrated Iris Xe)—development/prototyping
- **Cluster**: Remote server via SSH (Docker/K8s, multi-GPU planned)

SSH setup (`infra/deploy.sh`):

```
#!/bin/bash
ssh user@cluster-host << EOF
  docker pull ghcr.io/user/mlops:butterfly
  docker run -p 8080:8080 mlops:butterfly
EOF
```

CI/CD Mapping: Local tests on LoQ specs; cluster deploys via SSH-action.

3.5.4 Pipeline Validation Metrics

Success criteria:

- Accuracy: >88% top-1 val
- Energy: <0.5 kWh (CodeCarbon)
- Latency: <50ms inference (cluster)
- Overhead: Kepler pod energy <1.2x bare Docker

Butterfly simulates production CV serving while fitting resource constraints, benchmarking energy attribution accuracy central to this thesis.

3.6 Containerization and Kubernetes Deployment

Docker containers ensure reproducibility across local LoQ and SSH clusters, orchestrated via lightweight Kubernetes (k3s) with Kepler for energy observability.

3.6.1 Docker Setup

Per-use-case Dockerfile builds slim images:

```
FROM python:3.11-slim
COPY requirements.txt .
RUN pip install torch torchvision fastapi uvicorn
COPY src/ /app/src
CMD ["uvicorn", "app.main:app", "--host", "0.0.0.0"]
```

CI builds/pushes to GHCR; multi-stage reduces 2GB to 800MB.

3.6.2 Kubernetes (k3s) Cluster

Remote SSH cluster runs k3s (lightweight K8s):

```
curl -sfL https://get.k3s.io | sh -
kubectl apply -f monitoring/kepler.yaml
```

Deployment YAML structure:

- Deployment: 2 replicas, 2CPU/4Gi limits
- Service: Load balancer port 80→8080
- HPA: Auto-scale on CPU >70%

3.6.3 Kepler Energy Monitoring

Kepler DaemonSet deployment:

1. `kubectl apply -f kepler/kepler.yaml`
2. RAPL/NVML metrics via eBPF/cAdvisor
3. Prometheus export, Grafana dashboard

Core metric: `container_energy{pod='butterfly-vision'}`

3.6.4 Deployment Workflow

1. GitHub Actions: Docker build/push to GHCR
2. SSH-action: `kubectl apply -f k8s/`
3. Port-forward: `kubectl port-forward svc/butterfly 8080:80`
4. Monitor: Grafana kWh/pod realtime

Validates container overhead (10–20%) vs bare Docker, proving energy attribution accuracy.

List of Tables

List of Figures

References

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